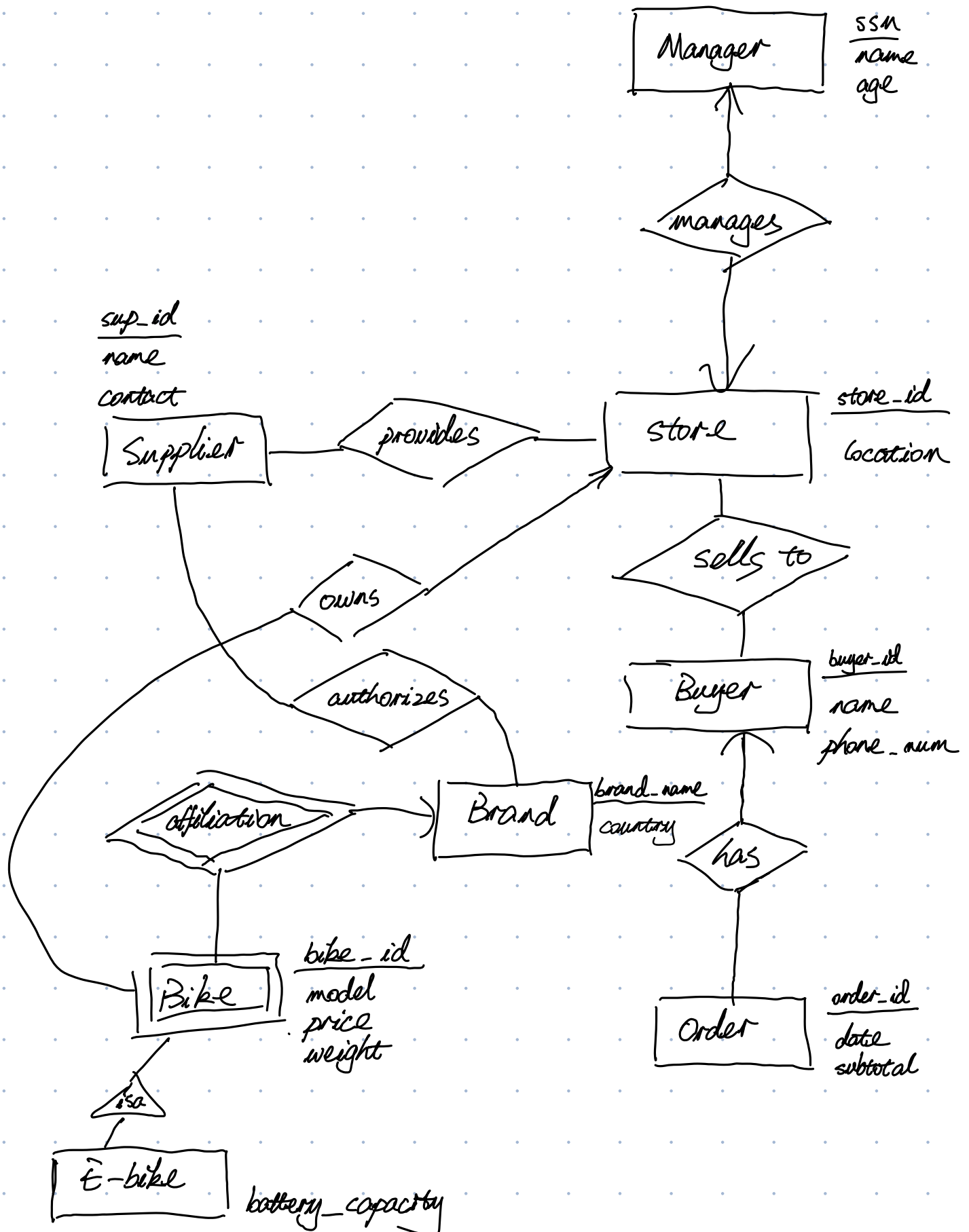


Ziqin Shen, Jie Wang, Xuecheng Sun.

We built a bike supply-chain database because all of us are bike-lovers.

We did all of the work together:

1. Brainstorming the entities and relations.
2. Building the ER diagram
3. Going to OT for verification of the ER diagram
4. Translating the ER diagram to table.
5. Going to OT again and editing our table.
6. Finishing this HW.



Supplier (sup-id, name, contact) Provides (sup-id, store-id) Store (store-id, location)

Authorizes (sup-id, brand-name) Brand (brand-name, country)

Manager (SSN, name, age, store-id)

Sells to (store-id, buyer-id) Buyer (buyer-id, name, phone-num)

Order (order-id, date, subtotal, buyer-id)

Bike (bike-id, model, price, weight, affiliated-brand, store-id)

E-Bike (bike-id, model, price, weight, affiliated-brand, battery-capacity, store-id)

We chose object-oriented method to translate this is-a relationship. All of the 3 methods have different advantages, so we chose the one according to our specific scenario.

We built a bike supply-chain database. We used object-oriented method because it reduces the computation power compared to ER style method. Battery capacity is a significant element that we take into consider when purchasing an e-bike. ER style method takes more time to look it up. Therefore, since we need to frequently look up this element, we will waste a lot of computational power. Also, our object-oriented method can save a lot of storage space. Since e-bikes takes only a small portion of bike's market, if we use NULLs, we will take up a lot of storage space for only storing NULLs. That's why we choose object-oriented method.